Reduced-Order Modelling of Dispersion

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Summary. We present low complexity models for the transport of passive scalars for environmental applications. Multi-level analysis has been used with a reduction in dimension of the solution space at each level. Similitude solutions are used in a non-symmetric metric for the transport over long distances. Model parameters identification is based on data assimilation. The approach does not require the solution of any PDE and, therefore, is mesh free. The model also permits to access the solution in one point without computing the solution over the whole domain. Sensitivity analysis is used for risk analysis and also for the identification of the sources of an observed pollution.

Key words: Reduced order modelling, source identification, risk analysis by sensitivity, non-symmetric geometry.

1 Introduction

Air and water contamination by pesticides is a major preoccupation for health and environment. One aims to model pesticide transport in atmospheric flows with very low calculation cost making assimilation-simulation and statistic risk analysis by Monte Carlo simulations realistic. In this problem available data is incomplete with large variability and the number of parameters involved large. Solution space reduction and reduced order modelling appear, therefore, as natural way to proceed.

Our contribution is to build a multi-level approach where a given level provides the inlet condition for the level above. In each level one aims to use a priori information in the definition of the search space for the solution and avoid the solution of partial differential equations.

More precisely, a near field (to the injection device) search space is build using experimental observations. Once this local solution known, the amount of specie leaving the atmospheric sub-layer is evaluated. This quantity is candidate for long distance transport using similitude solutions for mixing layers and plumes [Sim97]. These are known in Cartesian metrics. An original

contribution here is the generalization of these solutions in a non-symmetric travel-time based metric to account for non-uniform winds. We add constraint such that solutions built with this approach to be solution of the direct model (i.e. flow equations and transport model for a passive scalar). In particular, the divergence free condition for the generated winds, conservation, positivity and linearity of the solution of transport equations are requested.

Numerical examples show a comparison of this approach with a PDE based simulation. Examples also show multi-source configurations as well as sensitivity analysis of detected pollution. This is useful for both source identification and risk analysis.

2 Reduced-Order Modelling

One aims to model very large multi-scale phenomenon present in agricultural phyto treatment of cultures. The different entities to account for range from rows of plants to water attraction basins and one should also consider local topography and atmospheric conditions. It is, therefore, obvious that modelling phenomenon falling in length scales below a few meters becomes inevitable.

Consider the calculation of a state variable $V(p)$, function of independent variables p. Our aim is to define a suitable search space for the solution $V(p)$ instead of considering a general function space. This former approach is what one does in finite element methods, for instance, where the solution is expressed in some subspace $S(\{W_N\})$ described by the functional basis chosen $\{W_N\}$, with the quality of the solution being monitored either through the mesh quality and/or increasing the order of the finite element [Cia78]. In all cases, the size of the problem is large $1 \ll N < \infty$ and if the approach is consistent, the projected solution tends to the exact solution when $N \to \infty$.

In a low-complexity approach, one replaces the calculation of $V(p)$ by a projection over a subspace $S(\{w_n\})$ generated, for instance, by $\{w_n\}$, a family of solutions ('snapshots') of the initial full model ($p \to V(p)$). In particular, one aims $n \ll N$ [VP05].

In our approach, we aim to remove the calculation of these snapshots as this is not always an easy task. We take advantage of what we know on the physic of the problem and replace the direct model $p \to V(p)$ by an approximate model $p \to v(p)$ easier to evaluate. This is a very natural way to proceed, as often one does not need all the details on a given state. Also it is sufficient for the low-complexity model to have a local validation domain: one does not necessarily use the same low-complexity model over the whole range of the parameters. We have used this approach in the incomplete sensitivity concept where the linearization is performed not for the direct model but for an approximate state equation [MP01].